

CSS interview question

What is the full form of CSS?

Cascading style sheet

What does it mean by Cascading?

Cascading in CSS refers to the process by which the browser determines which style rules to apply when multiple rules target the same element. The term "cascading" highlights how CSS resolves conflicts between competing rules based on a hierarchy of priorities. This ensures that the most appropriate styles are applied.

Key Principles of Cascading:

1. Source of Styles:

- Styles can come from different sources: browser defaults, external stylesheets, inline styles, or user-defined styles.

2. Specificity:

- Rules with higher specificity (e.g., inline styles or more detailed selectors) take precedence over less specific ones.

3. Importance:

- Rules marked with !important override other rules, regardless of specificity.

4. Order of Appearance:

- When rules have the same specificity and importance, the one that appears last in the code takes precedence.

What is specificity?

CSS specificity is an algorithm that determines which style declaration is ultimately applied to an element.

If two or more CSS rules point to the same element, the declaration with the highest specificity will "win", and that style will be applied to the HTML element.

- Specificity is calculated based on the type of selectors used:
 - Inline styles (style="...") have the highest specificity.
 - IDs (#id) are more specific than classes (.class), attributes, or pseudo-classes.
 - Classes, attributes, and pseudo-classes are more specific than element selectors (div, p, etc.).
 - Universal selectors (*) and inherited styles have the lowest specificity.

What are pseudo-elements?

Pseudo-elements let you style **specific parts** of an element, even when that part doesn't actually exist in the HTML.

Examples:

- ::before
- ::after
- ::first-letter
- ::first-line

They help add decorative or structural content without extra HTML.

6. Do you know pseudo-classes?

Yes. Pseudo-classes represent a **state** of an element.

Examples:

- :hover → when mouse is over an element
- :active → when element is clicked
- :focus → when input is focused
- :checked → for checked checkboxes

They apply styles based on user interaction or element state.

7. What is the specificity of a pseudo-class and a pseudo-element?

Pseudo-class specificity = **same as a class**

- Specificity score: **0,1,0**

Pseudo-element specificity = **same as an element**

- Specificity score: **0,0,1**
-

8. What is the specificity of a normal class?

A normal class selector also has a specificity of:

0,1,0

Default value of position property?

Static

Difference between position relative and position absolute?

Absolute:

is another CSS technique that adjusts an element's position relative to its parent. If no parent element is present, the document body is used as the parent. The syntax for absolute positioning is **position: absolute;**

relative:

technique that allows an element to be adjusted from its normal position.

Fixed Position :is a CSS technique that keeps an element in the same place even when the page is scrolled. The syntax for fixed positioning is **position: fixed;** To position the element, we use top, right, bottom, and left properties.

What is flexbox?

It is layout system that we use in CSS for defining the layout of the webpage ex:
display : flex;

What is box model in CSS?

In CSS, the term "box model" is used when talking about web design and layout.

The CSS box model is essentially a box that wraps around every HTML element.

Every box consists of four parts: content, padding, borders and margins.

- **Content** - The content of the box, where text and images appear
- **Padding** - Clears an area around the content. The padding is transparent
- **Border** - A border that goes around the padding and content
- **Margin** - Clears an area outside the border. The margin is transparent

- **Outline in Box Model**

- The **outline** property in the CSS Box Model is used to draw a line around an element, outside the element's border. This property is useful for creating a visual focus or emphasis on an element. Unlike the border, the outline is not part of the element's dimensions, meaning it does not affect the element's size. The outline can be styled in various ways, such as solid, dashed, or dotted, and can have different colors. The outline is particularly useful for creating focus states or highlighting important elements on a webpage.

What do mean by box sizing in CSS? And what is its default value?

The CSS box-sizing property allows us to include the padding and border in an element's total width and height.

Its default value is content box

Difference between border box and content box?

- **content-box:** This is the default value of box-sizing. The dimension of element only includes 'height' and 'width' and does not include 'border' and 'padding' given to element. Padding and Border take space outside the element.

border-box: In this value, not only width and height properties are included but you will find padding and border inside of the box for example. `box {width: 200px; border: 10px solid black;}` renders a box that is 200px wide

Difference between inline, internal, and external CSS?

Inline targets a single element (highest specificity). Internal applies to the

current HTML file. External applies across pages and keeps styles centralized.

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Difference between inline and inline-block?

inline-block participates in line flow like inline elements but accepts width/height like block elements.

Explain CSS inheritance.

Some properties inherit (e.g., color, font-family); others do not (e.g., margin). The inherit and initial keywords explicitly control inheritance.

What is the display property?

Controls layout behavior: block, inline, inline-block, none, flex, grid, table, etc. display:none removes element from layout; visibility:hidden hides while keeping space.

What is float and why was it used?

float lets elements align left/right and text wrap around them (e.g., images). Historically used for layout, now largely replaced by Flexbox and Grid.

What is clear?

Prevents elements from wrapping around floated elements. Values: left, right, both, none.

CSS Grid Layout Module ?

The Grid Layout Module offers a grid-based layout system, with rows and columns.

The Grid Layout Module allows developers to easily create complex web layouts.

The Grid Layout Module makes it easy to design a responsive layout structure, without using [float](#) or positioning.

Q. What are Media Queries in CSS?

Media queries in CSS allows you to apply different styles to elements based on the size of device screen.

Media Query Syntax

A media query consists of an optional media-type and one or more media-features, which resolve to either true or false.

```
@media [not] media-type and (media-feature: value) and (media-feature: value) {  
    /* CSS rules to apply */  
}
```

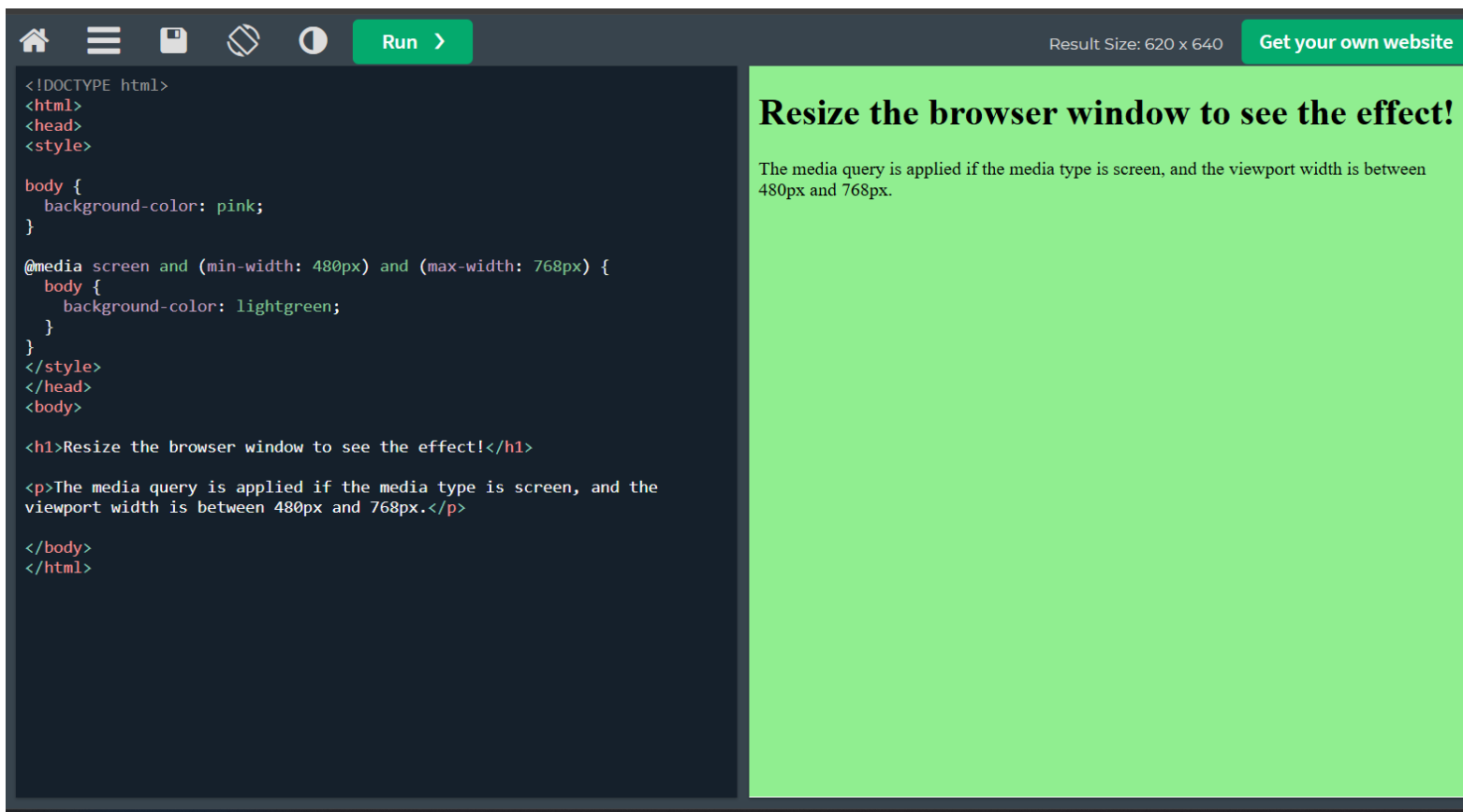
The media-type is optional. However, if you use **not**, you must also specify a media-type.

The result of a media query is true if the specified media-type matches the type of device, and all media-features are true. When a media query is true, the corresponding style rules are applied, following the normal cascading rules.

Meaning of the **not** and **and** keywords:

not: This optional keyword inverts the meaning of the entire media query.

and: This keyword combines a media-type and one or more media-features.



CSS Box Sizing ?

The CSS [box-sizing](#) property defines how to calculate the width and height of an element: should the calculation include padding and borders, or not.

By default, the width and height of an element is calculated like this:

- width + padding + border = actual width of an element
- height + padding + border = actual height of an element

border-box and ***content-box*** are the two different values of [box-sizing](#).

- **content-box:** This is the default value of box-sizing. The dimension of element only includes 'height' and 'width' and does not include 'border' and 'padding' given to element. Padding and Border take space outside the element.
- **border-box:** In this value, not only width and height properties are included but you will find padding and border inside of the box for example .box {width: 200px; border: 10px solid black;} renders a box that is 200px wide.

What are the 5 advantages of External Stylesheet:??

5 Advantages of CSS:

1. Separation of content(HTML) and presentation(Style).
2. Reusability of CSS classes in multiple elements.
3. Keeps things organized and structured.
4. Adapts to Different Devices.
5. Improves Website Speed.

Q. How to implement CSS using @import rule?

- ❖ @import rule is to include **external CSS** file to main CSS file.
- ❖ This method is less commonly used due to potential **performance issues**.

```
/* 3-Main.css */  
@import url("3-External.css");  
  
p {  
  font-size: 18px;  
}
```

```
/* 3-External.css */  
h1 {  
  color: blue;  
}
```

5 Ways of implementing CSS in HTML

1. Inline Styles
2. Internal Style Sheet
3. External Stylesheet
4. @import Rule
5. CSS Preprocessors

```
<html>  
  <head>  
    <link rel="stylesheet" href="3-Main.css" />  
  </head>  
  <body>  
    <h1>Imported Styles</h1>  
    <p>Paragraph</p>  
  </body>  
</html>
```


What are CSS Preprocessors? (SCSS)

CSS preprocessors are used to write styles in a special syntax that is then compiled into standard CSS. They extend the functionality of standard CSS by introducing features like **variables**, **nesting**, **mixins**, and **functions**. By using preprocessors, you can take advantage of advanced features that are not available in plain CSS.